

2013
**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 13, 17, 19, 23 and 25

Total marks – 100

Section I Pages 2–27

75 marks

This section has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this part

Part B – 55 marks

- Attempt Questions 21–31
- Allow about 1 hour and 40 minutes for this part

Section II Pages 29–39

25 marks

- Attempt ONE question from Questions 32–36
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 20 marks

Attempt Questions 1–20

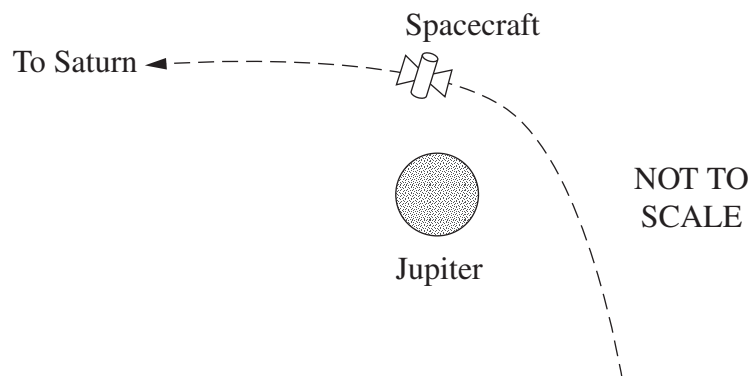
Allow about 35 minutes for this part

Use the multiple-choice answer sheet for Questions 1–20.

- 1 An investigation is designed to determine the size of the generated current when the strength of a magnet is varied.

Which is the independent variable for this investigation?

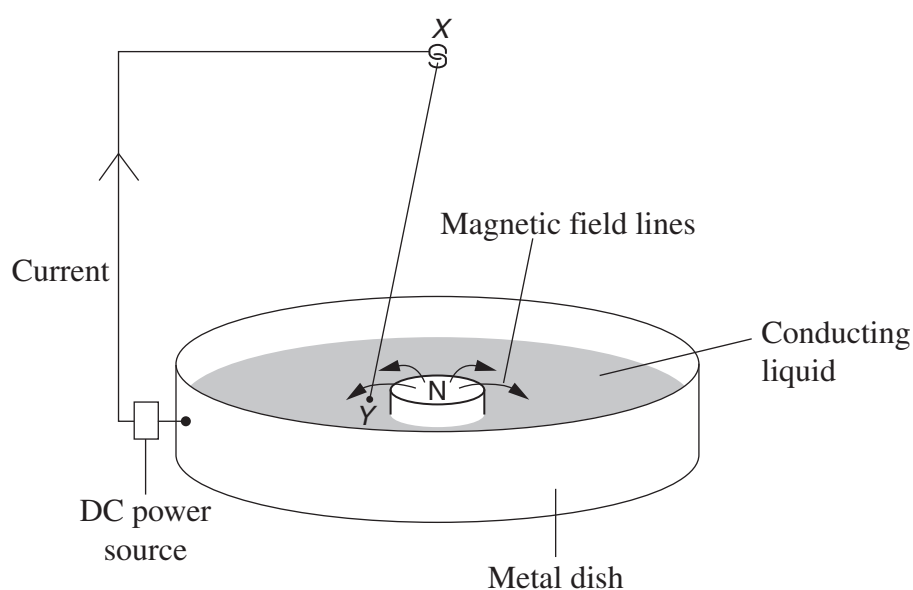
- (A) Speed of the magnet
 - (B) Strength of the magnet
 - (C) Size of the generated current
 - (D) Distance between the coil and the magnet
- 2 This diagram shows the path of a spacecraft as it goes past Jupiter and heads towards Saturn.



Which effect influences the path of the spacecraft?

- (A) Frictional
- (B) Motor
- (C) Photoelectric
- (D) Slingshot

- 3 The diagram shows equipment attached to a battery.



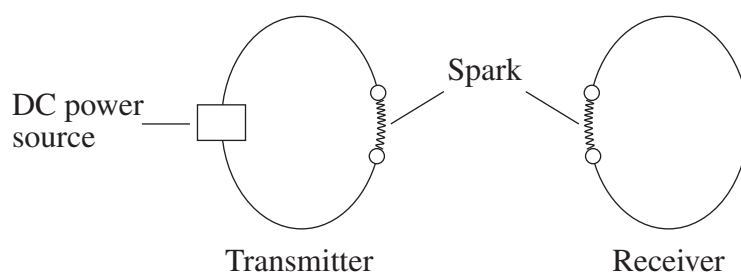
In which direction will the wire XY move?

- (A) Clockwise
 - (B) Anticlockwise
 - (C) Towards the magnet
 - (D) Away from the magnet
- 4 Students performed an investigation to determine the initial velocity of a projectile.

Which row correctly identifies a hazard of this investigation and a related precaution?

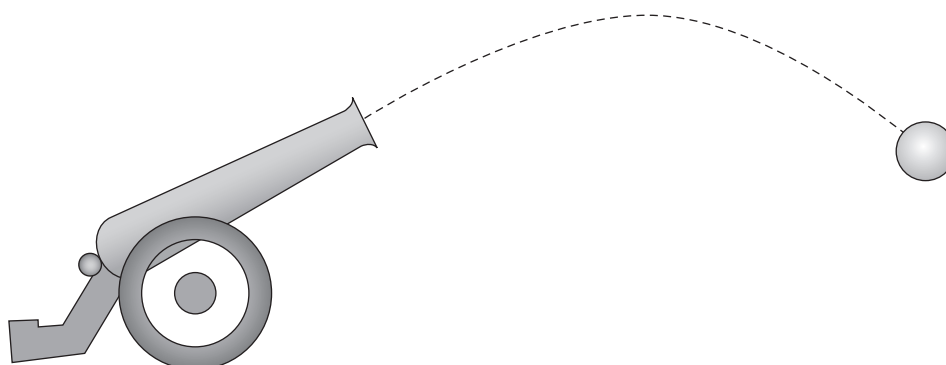
	<i>Hazard</i>	<i>Safety precaution</i>
(A)	flying projectile	wearing safety glasses
(B)	range of projectile	measuring the range with a tape measure
(C)	enclosed shoes	limiting the range of the projectile
(D)	safety glasses	flying projectile

- 5 The diagram represents an experiment that was conducted by Hertz.



What was Hertz trying to investigate?

- (A) Photoelectric effect
 - (B) Electrical induction
 - (C) Black body radiation
 - (D) Properties of radio waves
- 6 This diagram shows the path of a cannonball, fired from a cannon.



Which set of vectors represents the horizontal and vertical components of the cannonball's velocity along the path?

	<i>Horizontal</i>	<i>Vertical</i>
(A)	→ → → →	↓ ↓ ↓ ↓
(B)	→ → → →	↑ ↑ ↓ ↓
(C)	→ → → →	↑ ↑ ↓ ↓
(D)	→ → → →	↓ ↓ ↓ ↓

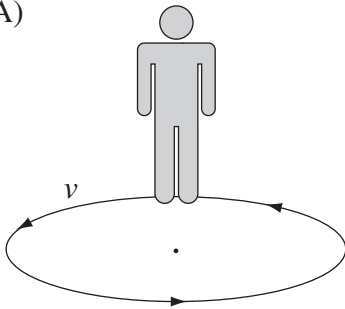
- 7 Eddy currents are a major source of energy loss in an iron core transformer.

What is one way to minimise this energy loss?

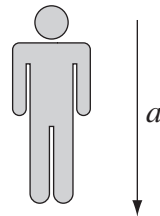
- (A) Laminate the iron core with an insulator
- (B) Put fewer turns of wire in the primary coil
- (C) Operate the transformer with a higher current
- (D) Decrease the distance between the primary and secondary coils

- 8 Which of the diagrams best represents an example of an inertial frame of reference?

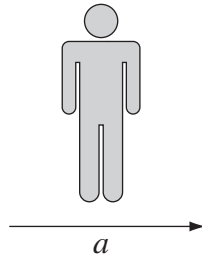
(A)



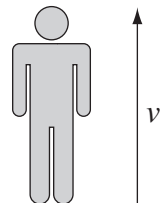
(B)



(C)



(D)



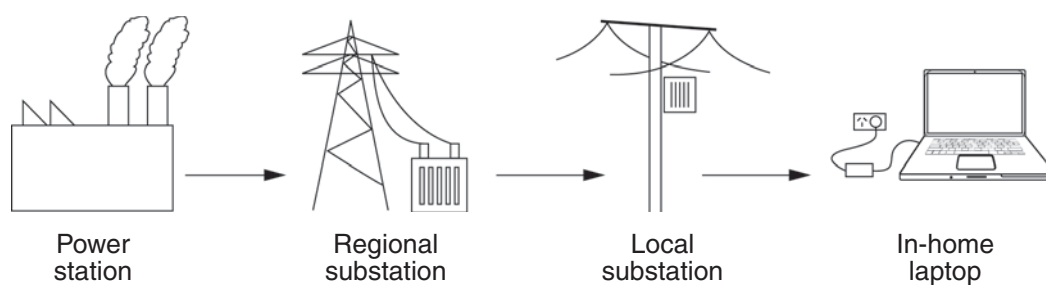
- 9 Three potentially hazardous conditions that can be experienced by astronauts in space are listed.

- I Extreme heat
- II Weightlessness
- III Communication blackout

Which combination of these conditions is associated with safe re-entry of a manned spacecraft?

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

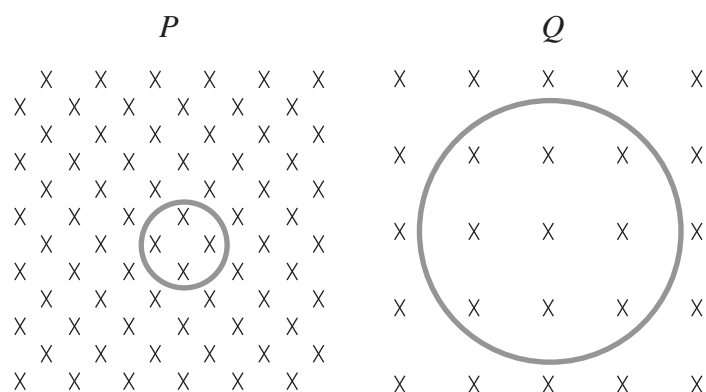
- 10 The diagram represents the transfer of electrical energy from a power station to a laptop computer.



Which flow diagram shows the correct use of transformers in this transfer?

- (A) Step-down → Step-up → Step-up → Step-down
- (B) Step-down → Step-down → Step-up → Step-up
- (C) Step-up → Step-down → Step-down → Step-down
- (D) Step-up → Step-up → Step-down → Step-up
- 11 Why did the Braggs use X-rays to determine the crystal structure of metals?
- (A) X-rays have a low frequency.
- (B) X-rays have a short wavelength.
- (C) X-rays can pass through metals.
- (D) X-rays travel at the speed of light.
- 12 Why is a magnet able to hover above a superconductor that is below its critical temperature?
- (A) The superconductor excludes magnetic flux.
- (B) The superconductor loses conductive properties when placed in a magnetic field.
- (C) The eddy current in the superconductor creates a potential difference between the magnet and the superconductor.
- (D) The gravitational force on the magnet is balanced by the force created by the crystal lattice structure of the superconductor.

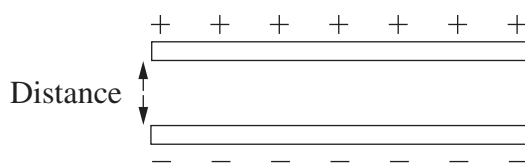
- 13 Different magnetic fields are passing through two copper rings, P and Q , as shown.



Which row of the table correctly identifies the ring with the greater magnetic flux and the ring with the greater magnetic flux density?

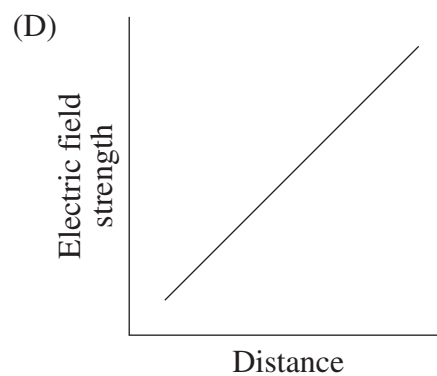
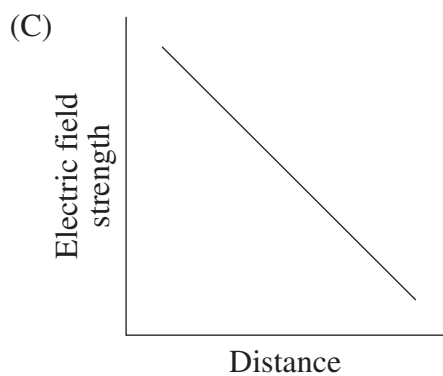
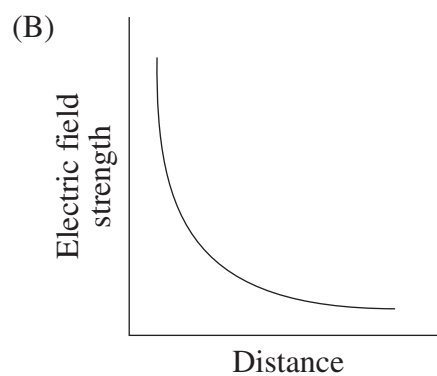
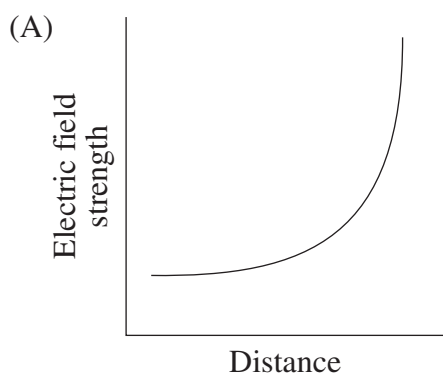
	<i>Greater magnetic flux</i>	<i>Greater magnetic flux density</i>
(A)	P	P
(B)	Q	Q
(C)	P	Q
(D)	Q	P

- 14 Two charged plates are initially separated by a distance as shown in the diagram.

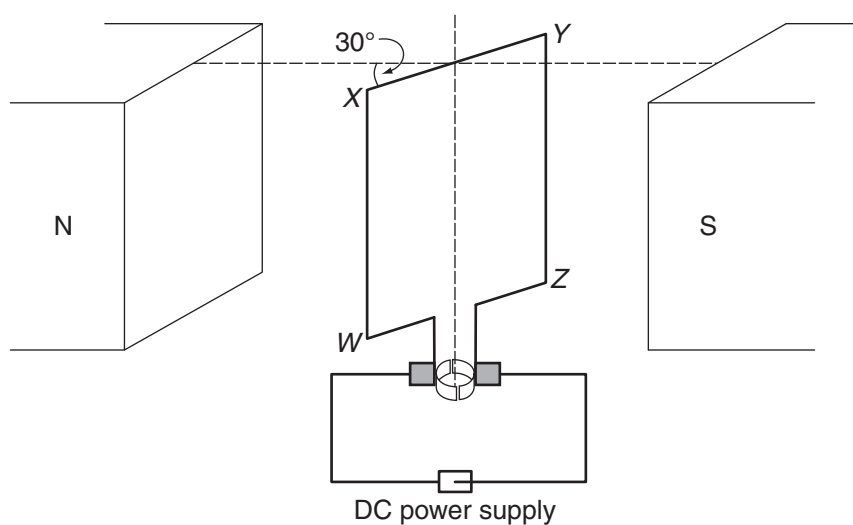


The potential difference between the plates remains constant.

Which of the graphs best represents the change in electric field strength as the distance between the two plates is increased?



- 15 The diagram shows a single-loop motor.



The equations $\tau = nBIA \cos\theta$ and $F = BIl \sin\omega$ can be used to calculate the torque in the motor and the force on the length of wire WX respectively.

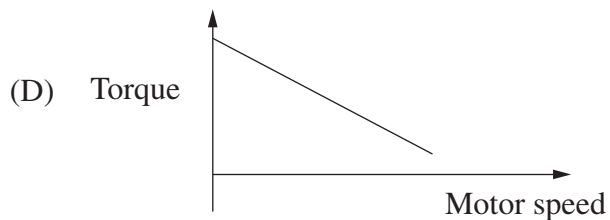
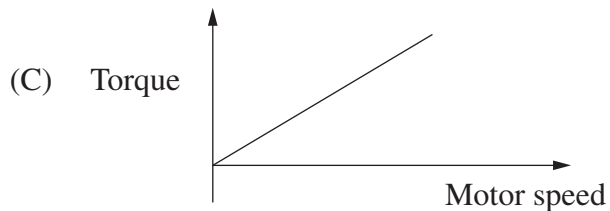
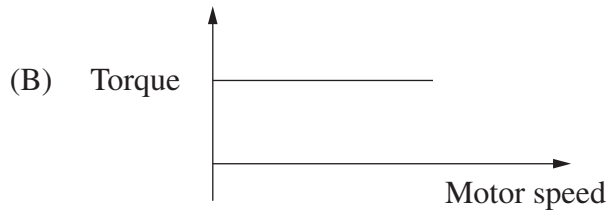
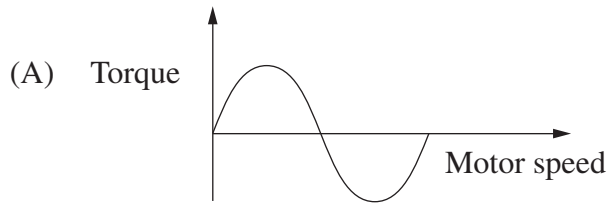
What angles are represented by θ and ω in the above equations?

	θ	ω
(A)	30°	90°
(B)	30°	30°
(C)	60°	90°
(D)	60°	30°

- 16 Which physical phenomenon allows current to flow through a metal wire?

- (A) Holes can move through a wire.
- (B) Charged atoms can move through a wire.
- (C) Electrons in a fully filled band can carry charge through a wire.
- (D) Electrons in a partially filled band can carry charge through a wire.

- 17 Which graph best represents the change in torque for a DC motor, with a radial magnetic field, from start up to operating speed?



- 18 Michelson and Morley conducted an experiment to measure the relative velocity of the Earth through the aether.

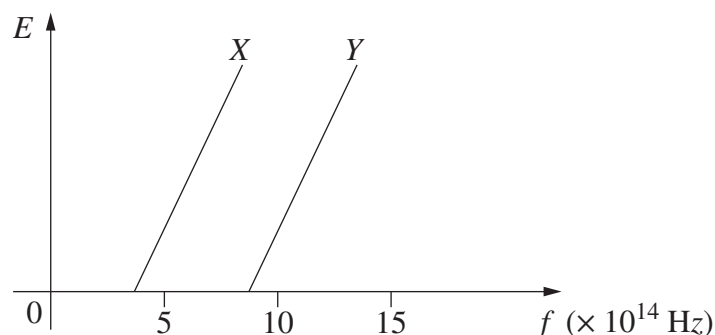
Why is the result of the experiment significant?

- (A) The result supported Michelson and Morley's hypothesis.
- (B) The result showed their hypothesis was incorrect because their investigation was not accurate enough.
- (C) The failure to detect the expected relative velocity contributed to the development of special relativity.
- (D) The successful verification of the expected relative velocity contributed to the development of special relativity.

- 19 A spaceship moves close to the speed of light, relative to a planet.

The rest-frame length of the spaceship can be determined by an observer who is

- (A) on the spaceship measuring the time taken for light to travel between two points on the planet.
 - (B) on the planet measuring the time taken for light to travel from the front to the back of the spaceship.
 - (C) on the spaceship measuring the time taken for light to travel from the front to the back of the spaceship.
 - (D) on the planet measuring the difference in the arrival time of light from the front and the back of the spaceship.
- 20 The graph shows the maximum kinetic energy (E) with which photoelectrons are emitted as a function of frequency (f) for two different metals X and Y .



The metals are illuminated with light of wavelength 450 nm.

What would be the effect of doubling the intensity of this light without changing the wavelength?

- (A) For metal X , the number of photoelectrons emitted would not change but the maximum kinetic energy would increase.
- (B) For metal X , the number of photoelectrons emitted would increase but the maximum kinetic energy would remain unchanged.
- (C) For both metals X and Y , the number of photoelectrons emitted would not change but the maximum kinetic energy would increase.
- (D) For both metals X and Y , the number of photoelectrons emitted would increase but the maximum kinetic energy would remain unchanged.

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Centre Number

Section I (continued)

Part B – 55 marks

Attempt Questions 21–31

Allow about 1 hour and 40 minutes for this part

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Student Number

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Question 21 (4 marks)

During the course you studied one of the following scientists who has contributed to the development of space exploration:

- Tsiolkovsky
- Esnault-Pelterie
- Oberth
- O'Neill
- Goddard
- von Braun

Name of scientist studied:

- (a) Outline ONE contribution made by the named scientist to the development of space exploration. 2

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- (b) Outline how to ensure that secondary sources of information about the scientist are reliable. 2

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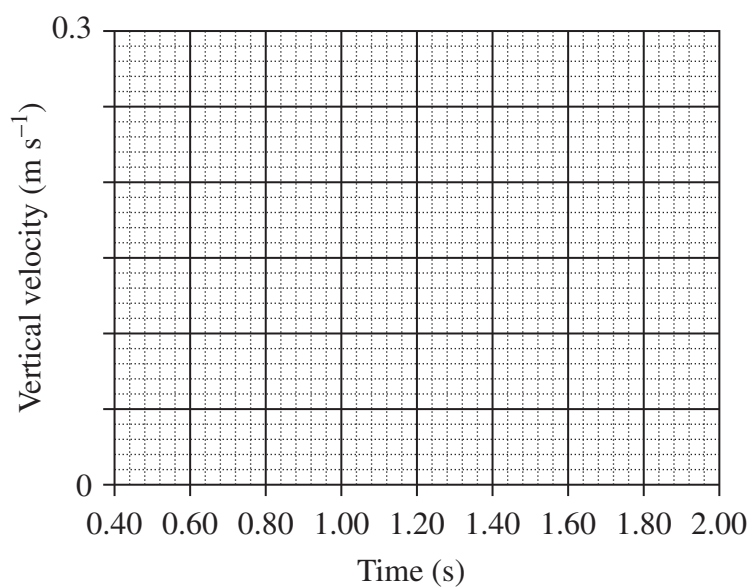
Question 22 (3 marks)

This set of data was obtained from a motion investigation to determine the acceleration due to gravity on a planet other than Earth.

3

Time (s)	Vertical velocity (m s^{-1})
0.60	0.02
1.00	0.09
1.20	0.12
1.40	0.17
1.80	0.23

Plot the data from the table, and then calculate the acceleration.



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Question 23 (5 marks)

A planet orbits the star, Pollux, at a distance of 1.64 astronomical units (AU). It takes 590 Earth days to complete one orbit.

- (a) Why does the mass of the planet play NO role in determining its orbital speed around Pollux? **2**

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- (b) A satellite orbits Pollux with a period of 365 Earth days. **3**

How far is the satellite from Pollux in astronomical units (AU)?

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2013 HIGHER SCHOOL CERTIFICATE EXAMINATION
Physics

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Centre Number

Section I – Part B (continued)

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Question 24 (5 marks)

With the aid of a diagram, explain how a photocell works. 5

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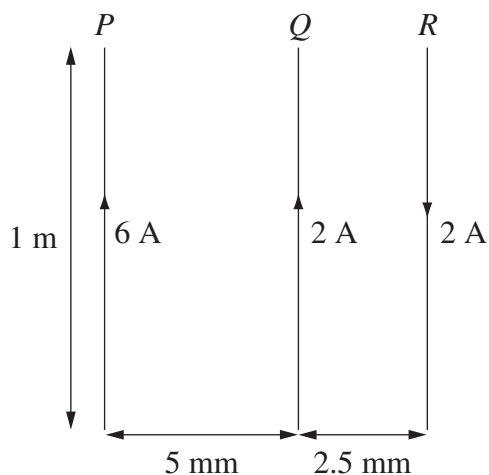
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Question 25 (4 marks)

P , Q and R are straight current-carrying conductors.

Conductors P and R are fixed and unable to move. Conductor Q is free to move.



- (a) In which direction will the conductor Q move as a result of the current flow in P and R ? **1**

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- (b) Calculate the magnitude of the force experienced by Q as a result of the currents through P and R . **3**

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Centre Number

Section I – Part B (continued)

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Student Number

Question 26 (5 marks)

An electric field is produced between two charged parallel plates, M and N .

M

N

- (a) The plates, M and N , are 1.0 cm apart and have an electric field of 15 V m^{-1} . 2

Calculate the potential difference between the plates.

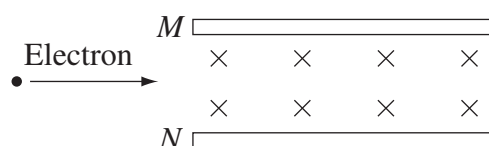
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- (b) The potential difference is now changed and a magnetic field of 0.5 T is placed perpendicular to the plates, as shown in the diagram below. 3



Determine the magnitude and direction of the electric field required to allow the electron to travel through undeflected, if the electron is moving at $1 \times 10^4 \text{ m s}^{-1}$.

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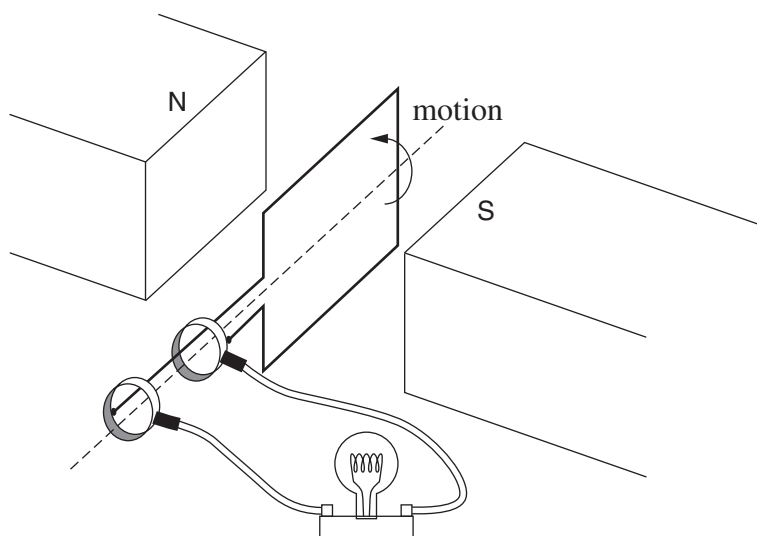
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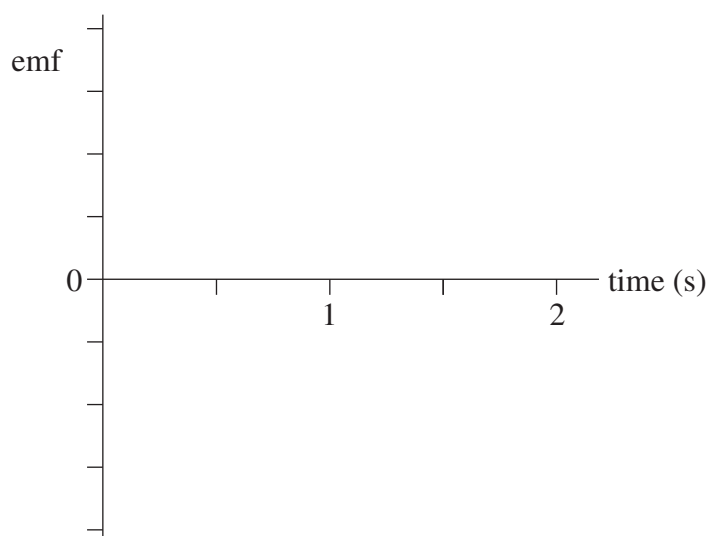
Question 27 (7 marks)

- (a) A generator starts at the position shown and is rotated by one revolution in the first second. It is then rotated by two revolutions in the next second.

3



Sketch a graph on the axes showing the electromotive force (emf) produced by this generator for these two seconds.



Question 27 continues on page 21

Question 27 (continued)

- (b) An electric motor can be used to propel a vehicle. The electric motor can be made to operate as a generator when the vehicle is moving. This will have a braking effect on the vehicle.

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Explain the physics principles involved in the propelling and braking of this vehicle.

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End of Question 27

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Centre Number

Section I – Part B (continued)

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Student Number

Question 28 (6 marks)

Assess Einstein's contribution to quantum theory.

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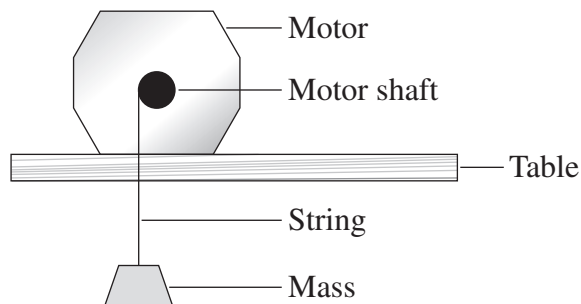
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Question 29 (5 marks)

A 0.05 kg mass is lifted at a constant speed by a DC motor. The motor has a coil of 100 turns in a 0.1 T magnetic field. The area of the coil is 0.0012 m^2 . The motor shaft has a radius of 0.004 m.



- (a) Determine the force needed to lift the mass. 2

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- (b) Calculate the minimum current required in the coil to lift the mass. 3

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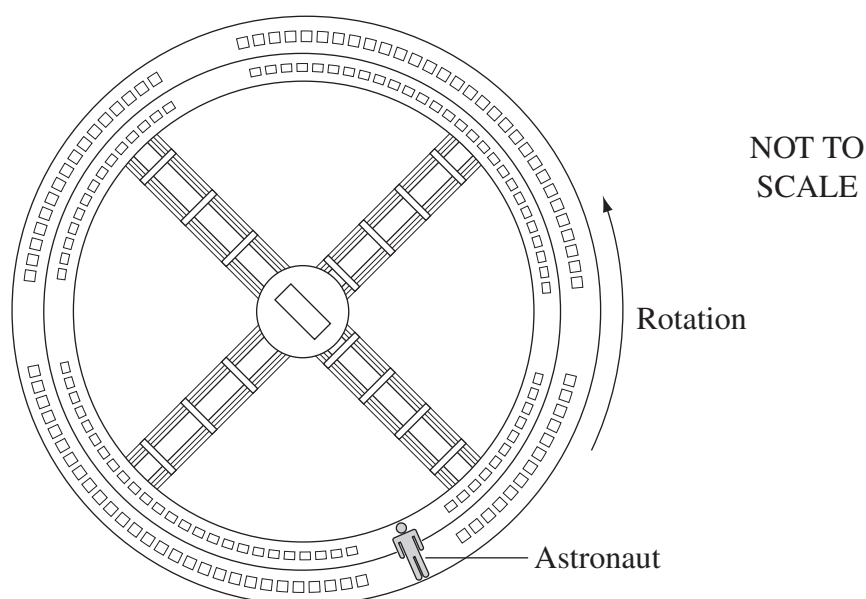
Section I – Part B (continued)

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Student Number

Question 30 (4 marks)

The diagram shows a futuristic space station designed to simulate gravity in a weightless environment.



- (a) Explain how rotating the space station simulates gravity for the astronaut.

2

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Question 30 continues on page 26

Question 30 (continued)

- (b) Calculate the rotational speed that a space station with a diameter of 550 m would need for astronauts to experience 1 *g* of acceleration. **2**

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End of Question 30

Question 31 (7 marks)

Explain how changing the electrical properties of materials has led to the development of new technologies which can benefit society.

7

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Physics

Section II

25 marks

Attempt ONE question from Questions 32–36

Allow about 45 minutes for this section

For Question 32

- answer parts (a)–(c) of the question in Section II Answer Booklet 1.
- answer parts (d)–(f) of the question in Section II Answer Booklet 2.

Extra writing booklets are available.

For Questions 33, 34 and 36

- answer parts (a)–(d) of the question in Section II Answer Booklet 1.
- answer parts (e)–(f) of the question in Section II Answer Booklet 2.

Extra writing booklets are available.

For Question 35

- answer parts (a)–(c) of the question in Section II Answer Booklet 1.
- answer parts (d)–(e) of the question in Section II Answer Booklet 2.

Extra writing booklets are available.

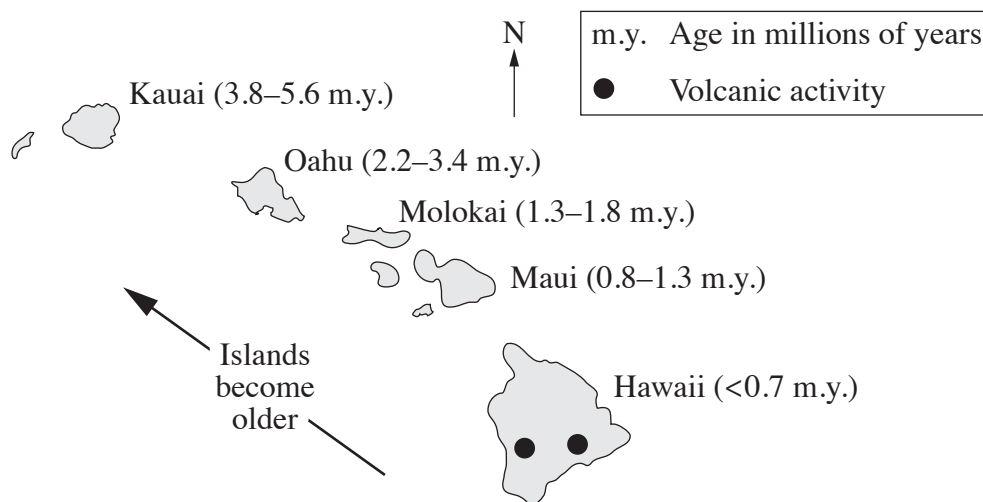
Show all relevant working in questions involving calculations.

	Pages
Question 32 Geophysics	30–31
Question 33 Medical Physics	32–33
Question 34 Astrophysics	34–35
Question 35 From Quanta to Quarks	36
Question 36 The Age of Silicon	37–39

Question 32 — Geophysics (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) Describe how ONE piece of geophysics equipment detects seismic waves. **3**
- (b) (i) Describe a procedure that could be followed to determine the density of rock types. **3**
- (ii) How could the reliability of the results of part (b) (i) be determined? **2**
- (c) The diagram shows the age and position of each island in the Hawaiian island chain.



Acknowledgement: © Australian Academy of Science

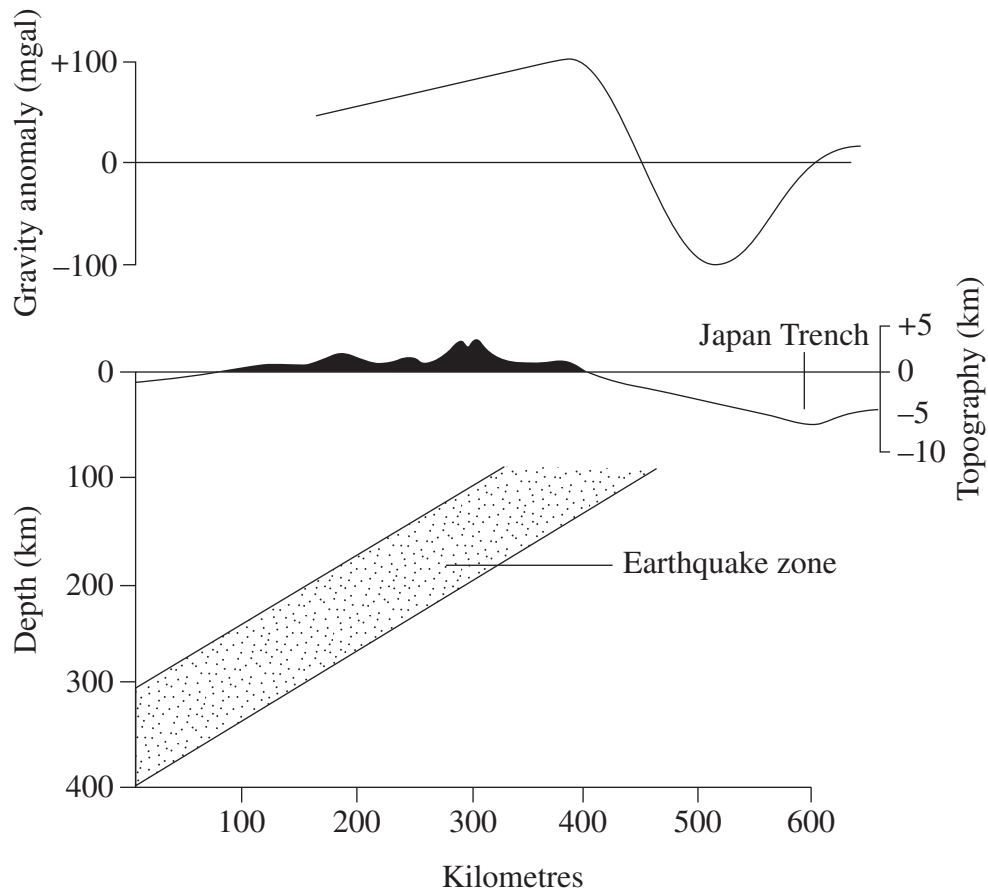
- (i) Explain the volcanic activity on the island of Hawaii. **2**
- (ii) Explain how the formation of the Hawaiian island chain can be used as evidence for the theory of plate tectonics. **3**

Question 32 continues on page 31

Question 32 (continued)

Answer parts (d)–(f) in Section II Answer Booklet 2.

- (d) The graphs show the observed gravity anomaly across the Japan Trench and its relationship to topography. The earthquake zone is dipping away from the trench. 3



Acknowledgement: © Australian Academy of Science

Explain the significance of the shape of the gravity anomaly line with reference to the graphs.

- (e) The Earth has a radius of 6357 km through the poles. Show how the acceleration due to gravity at the North Pole can be determined. 3
- (f) Discuss the effectiveness of using geophysical methods in mineral exploration and environmental monitoring. 6

End of Question 32

Question 33 — Medical Physics (25 marks)

Answer parts (a)–(d) in Section II Answer Booklet 1.

- | | | |
|-----|--|----------|
| (a) | Outline differences between X-ray images and CAT scans. | 3 |
| (b) | Outline a first-hand investigation to demonstrate the transfer of light by optical fibres. | 2 |
| (c) | Explain how an endoscope can be used to obtain tissue samples of internal organs. | 3 |
| (d) | (i) How is the orientation of the magnetic axis of a hydrogen nucleus affected by the application of a strong magnetic field during an MRI scan? | 2 |
| | (ii) Explain the difference in the relaxation time of hydrogen in water and the relaxation time of hydrogen in other molecules. | 3 |

Question 33 continues on page 33

Question 33 (continued)

Answer parts (e)–(f) in Section II Answer Booklet 2.

- (e) (i) How is the Doppler effect used in ultrasonics to investigate blood flow characteristics? **3**
- (ii) Outline TWO technologies that are required for an ultrasound scan. **3**
- (f) The data set gives information about a variety of radioactive isotopes, some of which can be used in diagnostic tools for medicine. **6**

<i>Radioactive isotope</i>	<i>Radiation emitted</i>	<i>Half life</i>
Fluorine-18	β^+	109.77 minutes
Strontium-90	β	28.8 years
Carbon-14	β	5730 years
Technetium-99m	γ	6 hours
Cobalt-60	γ	5.27 years

Using the information in the table, justify which radioactive isotopes are suitable for use in diagnostic imaging techniques.

End of Question 33

Question 34 — Astrophysics (25 marks)

Answer parts (a)–(d) in Section II Answer Booklet 1.

- (a) Describe an investigation that can be used to show why it is desirable in terms of sensitivity for a telescope to have a larger diameter lens. **3**

- (b) Matar is a star in the same spectral class as the Sun, yet it has four times the mass of the Sun.
 - (i) Draw a labelled Hertzsprung–Russell (H–R) diagram, and mark the position of Matar on the diagram with an X. **3**
 - (ii) Contrast the physical properties, other than mass, of Matar and the Sun. **2**

- (c) Describe the dominant nuclear reaction in an A2 main sequence star. **3**

- (d) Describe the transition process of a G2 main sequence star, such as the Sun, evolving into a red giant. **3**

Question 34 continues on page 35

Question 34 (continued)

Answer parts (e)–(f) in Section II Answer Booklet 2.

- (e) The graphs give information that could be used to calculate the distance from Earth to a Type 1 Cepheid Variable.

Awaiting copyright

- (i) The star is claimed to be approximately 1260 parsecs from Earth. **3**

Show how this claim is supported or refuted by the information in the graphs.

- (ii) State an advantage and a disadvantage of using the graphs above to calculate the distance to a star. **2**

- (f) How is light analysed, using physics principles, in order to determine the standard properties for the main sequence stars of Spectral Class A shown in this table? **6**

Spectral Class	Effective Temperature (K)	Colour	Mass/Mass _{Sun}	Luminosity/Luminosity _{Sun}
A	7500 – 10 000	White	2.0 – 3.0	8 – 55

Acknowledgement: CSIRO

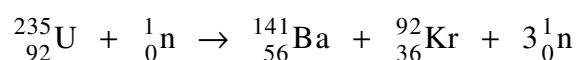
End of Question 34

Question 35 — From Quanta to Quarks (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) (i) Outline the key features of the Rutherford model of the atom. **2**
- (ii) Explain how the Bohr model of the atom overcomes an identified limitation of the Rutherford model of the atom. **3**

- (b) This equation describes a nuclear reaction.



Particle	Mass (amu)
${}_{92}^{235}\text{U}$	235.0439
${}_{56}^{141}\text{Ba}$	140.9144
${}_{36}^{92}\text{Kr}$	91.9263
${}_0^1\text{n}$	1.008 665

- (i) Use the data above to calculate the energy released in this reaction. **3**
- (ii) Apply Einstein's idea of the equivalence of mass and energy to explain the production of energy in the reaction above. **2**
- (c) Pauli is said to have 'suggested' the existence of the neutrino. **4**

Explain why the neutrino was 'suggested' but not discovered by Pauli.

Answer parts (d)–(e) in Section II Answer Booklet 2.

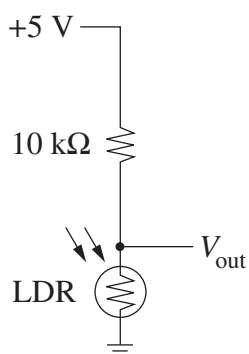
- (d) (i) Identify the fundamental forces proposed by the standard model of the atom. **2**
- (ii) Explain how the stability of atomic nuclei is maintained. **3**
- (e) How did de Broglie use existing concepts and ideas to come up with new interpretations that have increased our understanding of the structure of matter? **6**

Question 36 — The Age of Silicon (25 marks)

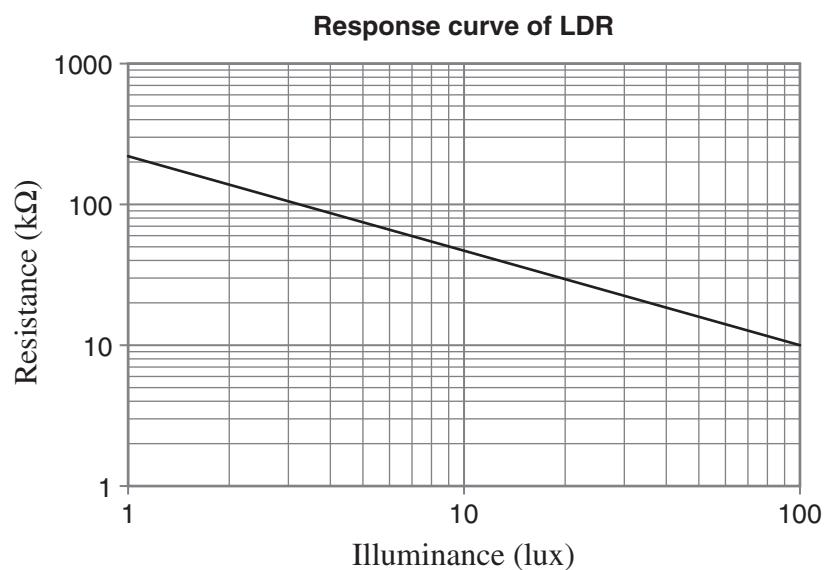
Answer parts (a)–(d) in Section II Answer Booklet 1.

- (a) This circuit shows how a light-dependent resistor (LDR) can be used to control a voltage output.

3



The response curve of the LDR in the circuit is shown.

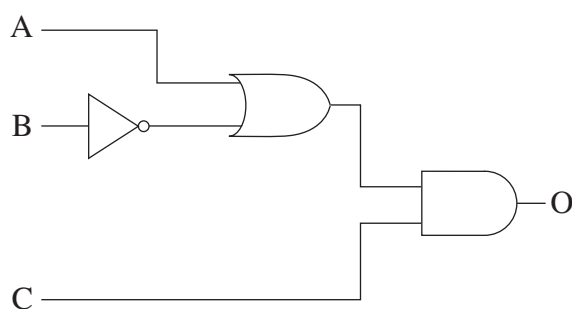


What would the voltage V_{out} be when the light intensity at the LDR is 20 lux?

Question 36 continues on page 38

Question 36 (continued)

- (b) Describe the role of a LDR in a camera. **3**
- (c) Explain how input and output transducers can be used to control the temperature in a room. **4**
- (d) For the logic circuit, construct a truth table showing the output, O, for all possible combinations of inputs A, B and C. **3**



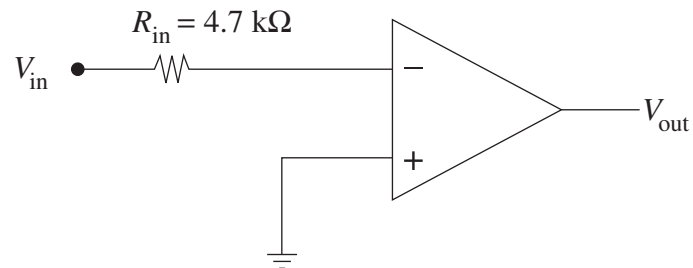
Question 36 continues on page 39

Question 36 (continued)

Answer parts (e)–(f) in Section II Answer Booklet 2.

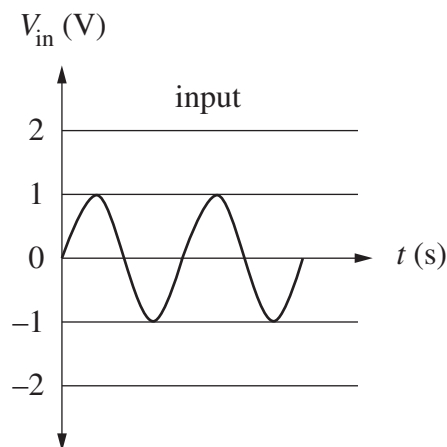
(e) (i) Describe an ideal operational amplifier. 2

(ii) The diagram shows an operational amplifier. 2



Reproduce this diagram in your writing booklet and modify it to perform as an inverting amplifier with a gain of 2.

(iii) The diagram shows an input waveform V_{in} for an inverting amplifier with a gain of 2. 2



Sketch the output waveform V_{out} in your writing booklet.

(f) Assess how the development of electronics since the invention of the transistor has met the increasing computing needs of society. 6

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DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi} \right)$	$2.0 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{\text{av}} = \frac{\Delta r}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} \text{ therefore } a_{\text{av}} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2}a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = - \frac{R_{\text{f}}}{R_{\text{i}}}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY										2 He 4.003 Helium					
3 Li 6.941 Lithium	4 Be 9.012 Beryllium	Atomic Number Symbol		Standard Atomic Weight Name								9 F 19.00 Fluorine					
		79 Au 197.0 Gold															
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium																
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.38 Zinc	31 Ga 69.72 Gallium	32 Ge 72.64 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton
37 Rb 85.47 Rubidium	38 Sr 87.61 Strontium	39 Y 88.91 Yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.96 Molybdenum	43 Tc Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon
55 Cs 132.9 Caesium	56 Ba 137.3 Barium	57–71 Lanthanoids		72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.9 Tungsten	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po Polonium	85 At Astatine	86 Rn Radon
87 Fr Francium	88 Ra Radium	Actinoids	Rutherfordium	Dubnium	Seaborgium	Bohrium	Hassium	Meitnerium	Darmstadtium	Roentgenium	Copernicium						
Lanthanoids																	
57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium			
Actinoids																	
89 Ac Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium			

Elements with atomic numbers 113 and above have been reported but not fully authenticated.
Standard atomic weights are abridged to four significant figures.
Elements with no reported values in the table have no stable nuclides.
The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of data. Some data may have been modified.



2013 HSC Physics Marking Guidelines

Section I, Part A

Multiple-choice Answer Key

Question	Answer
1	B
2	D
3	A
4	A
5	D
6	C
7	A
8	D
9	B
10	C
11	B
12	A
13	D
14	B
15	A
16	D
17	D
18	C
19	C
20	B

Section I, Part B

Question 21 (a)

Criteria	Marks
• Provides an outline of one contribution of the named scientist to the development of space exploration	2
• Identifies a contribution of one of the listed scientists	1

Sample answer:

von Braun	Launched the V2 rocket demonstrating the use of gyroscopes to stabilise large rockets.
-----------	--

Answers could include:

Scientist	Contribution
Tsiolkovsky	Calculated the speed a rocket would need to escape the Earth's atmosphere.
Oberth	Theorist who promoted interest in rocketry, developed equations for spaceflight.
Goddard	He launched the world's first liquid fuelled rocket, using alcohol and liquid oxygen to reduce the overall mass of the rocket.
Esnault-Pelterie	Experimented with various liquid fuel mixes.
O'Neill	Suggested the idea of space stations at Lagrange points.

Question 21 (b)

Criteria	Marks
• Outlines a valid method to ensure reliability of secondary sources of information about the scientist	2
• Shows some understanding of reliability	1

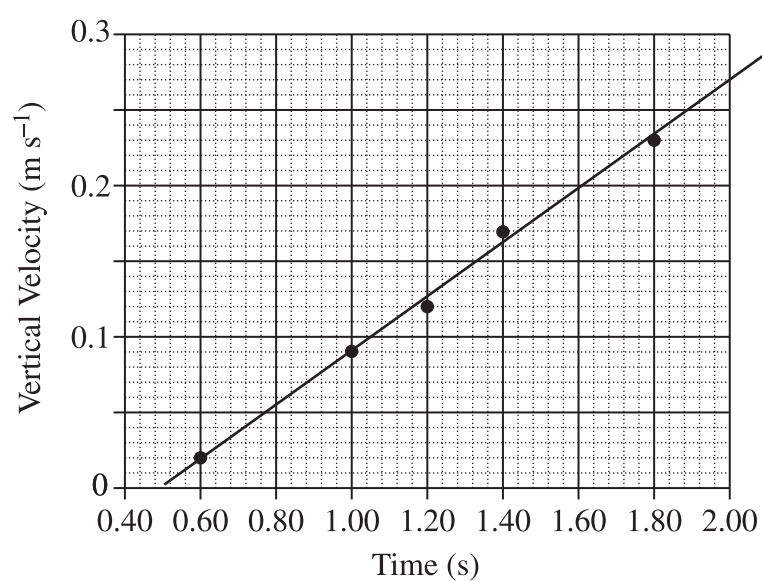
Sample answer:

A source could be cross-referenced with several other reputable sources. If the information is consistent it is deemed reliable.

Question 22

Criteria	Marks
- Correctly plots points and draws line of best fit - Correct substitution to determine the acceleration	3
- Correctly plots points and attempts to determine the acceleration OR - Correctly plots points and draws line of best fit	2
Shows a basic understanding of graphing	1

Sample answer



Acceleration

$$= \frac{0.235 - 0.02}{1.80 - 0.60}$$

$$= \frac{0.215}{1.20}$$

$$= 0.18 \text{ m s}^{-2}$$

Question 23 (a)

Criteria	Marks
Explains why the mass of the planet plays no role in determining its orbital speed	2
Shows some understanding of the orbital speed of the planet	1

Sample answer:

The orbital velocity is determined by the planet's period and distance from the central mass, as described by Kepler's Laws. Therefore, the mass of the planet plays no role in determining its orbital speed around Pollux.

Question 23 (b)

Criteria	Marks
Demonstrates correct process to calculate the distance required	3
Relates period and radius of the 2 objects orbiting the central mass OR	2
Substitutes into correct equation to calculate constant	
Partial substitution into a relevant formula OR	1
Attempts to relate period and radius of the 2 objects orbiting the central mass	

Sample answer:

$$\frac{r_x^3}{T_x^2} = \frac{r^3}{T^2} \quad \frac{r^3}{365^2} = 1.27 \times 10^{-5}$$

$$\frac{1.64^3}{590^2} = \frac{r^3}{365^2} \quad r^3 = (1.27 \times 10^{-5}) \times 365^2$$

$$r = \sqrt[3]{(1.27 \times 10^{-5}) \times 365^2}$$

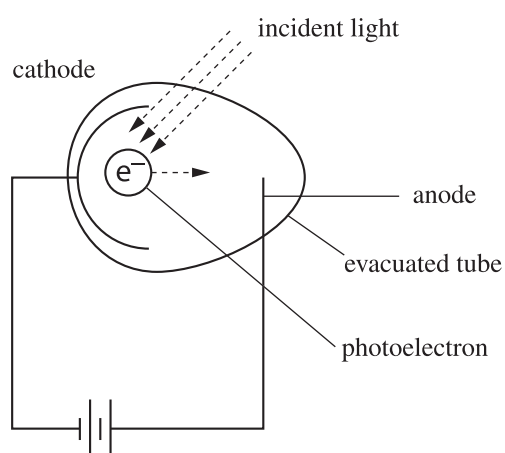
$$= 1.19 \text{ AU}$$

Question 24

Criteria	Marks
- Explains how a photocell works, using photoelectric effect - Includes an appropriate diagram	5
Describes how a photocell works, using photoelectric effect	4
Outlines how a photocell works, using photoelectric effect	3
Outlines how a photocell works, without necessarily mentioning the photoelectric effect	2
- Identifies ONE feature of a cell that uses light OR - Identifies that a photocell uses the photoelectric effect	1

Sample answer:

A photocell detects light and transforms the energy into electricity using the photoelectric effect. When light of frequency greater than the threshold frequency of the cathode material falls on the cathode, photoelectrons are emitted because the electrons have enough energy to leave the surface of the metal. This is the photoelectric effect. These electrons are collected by the anode and an electric current starts flowing in the external circuit. The current would stop if the light does not fall on the cathode or the frequency of the light was less than the threshold frequency.



Question 25 (a)

Criteria	Marks
• Identifies correct direction	1

Sample answer:

To the left

OR

Towards the conductor P

Question 25 (b)

Criteria	Marks
• Demonstrates correct process to calculate the force experienced by Q	3
• Demonstrates logical process to calculate force P or force R on Q	2
• Partial substitution into a relevant equation	1

Sample answer:

$$\frac{F}{l} = \frac{kI_1 I_2}{d}$$

Force P on Q

$$F = \frac{2 \times 10^{-7} \times 6 \times 2}{5 \times 10^{-3}}$$

$$F = 4.8 \times 10^{-4} \text{ N}$$

Force R on Q

$$F = \frac{2 \times 10^{-7} \times 2 \times 2}{2.5 \times 10^{-3}}$$

$$F = 3.2 \times 10^{-4} \text{ N}$$

$$\text{Total force} = 4.8 \times 10^{-4} \text{ N} + 3.2 \times 10^{-4} \text{ N}$$

$$\text{Total force} = 8 \times 10^{-4} \text{ Newtons}$$

Question 26 (a)

Criteria	Marks
Substitutes into correct equation to calculate the potential difference between the plates	2
Partial substitution into a relevant equation	1

Sample answer:

$$E = \frac{V}{d}$$

$$\therefore V = Ed = 15 \times 1.0 \times 10^{-2} = 0.15 \text{ V}$$

Question 26 (b)

Criteria	Marks
- Demonstrates correct process to determine the magnitude of the electric field AND - Identifies the correct direction of the electric field	3
- Demonstrates correct process to determine the magnitude of the electric field OR - Correctly substitutes into a relevant equation and identifies the correct direction of the electric field	2
Identifies ONE relevant piece of data	1

Sample answer:

$$F = qvB = qE$$

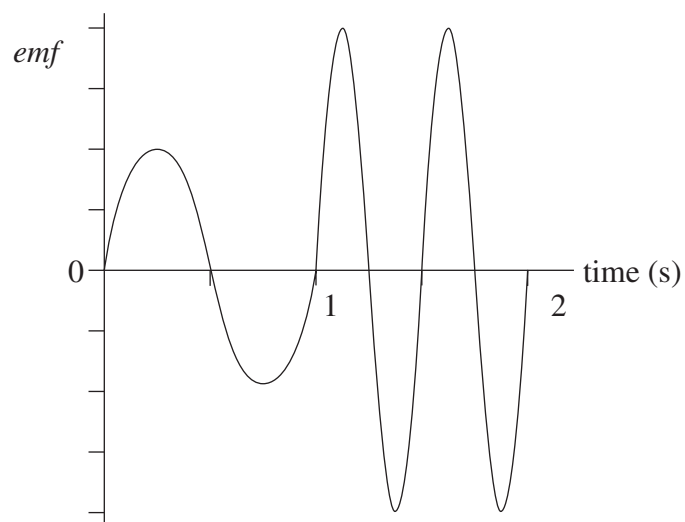
$$\therefore E = vB = 1 \times 10^4 \times 0.5$$

$$= 5000 \text{ V m}^{-1} \text{ from } M \text{ to } N$$

Question 27 (a)

Criteria	Marks
Starting from zero, produces a single period AC curve for 0–1 seconds, followed by an AC curve of twice the frequency and amplitude for 1–2 seconds	3
- Produces a correct curve for 0–1 seconds OR 1–2 seconds AND - Includes a correct feature for the other second	2
Identifies ONE correct feature	1

Sample answer:



Question 27 (b)

Criteria	Marks
• Demonstrates a sound understanding of the physics principles involved in a motor and a generator • Considers the energy changes and relates these to the motion of the vehicle both in propulsion and in braking • Clearly explains the physics principles involved in the propelling and braking of the vehicle	4
• Demonstrates an understanding of the physics principles involved in a motor and a generator • Describes relevant energy changes	3
• Demonstrates an understanding of the physics principles involved in a motor or generator OR • Shows some understanding of how a motor can act as a generator	2
• Identifies some relevant information	1

Sample answer:

A motor consists of a rotating coil in a magnetic field. When power is cut, this rotation causes an emf to be produced due to a change in magnetic flux. This allows the motor to act as a generator, which has essentially the same parts as a motor.

When acting as a motor, the vehicle converts electrical energy to kinetic energy due to the motor effect, thus propelling the vehicle. When it acts as a generator, kinetic energy is converted to electrical energy, and so by Lenz's law, the motion of the rotor is opposed. This acts to slow the vehicle.

Question 28

Criteria	Marks
- Demonstrates a thorough understanding of the contribution of Einstein to quantum theory - Makes an informed judgement implicitly or explicitly about Einstein's contribution to quantum theory	6
- Demonstrates a thorough understanding of the contribution of Einstein to quantum theory OR - Demonstrates a good understanding of the contribution of Einstein to quantum theory AND - Makes an informed judgement implicitly or explicitly about Einstein's contribution to quantum theory	5
- Demonstrates a good understanding of the contribution of Einstein to quantum theory OR - Demonstrates a sound understanding of the contribution of Einstein to quantum theory AND - Makes an informed judgement implicitly or explicitly about Einstein's contribution to quantum theory	4
- Demonstrates a sound understanding of the contribution of Einstein to quantum theory - OR - Demonstrates some understanding of the contribution of Einstein to quantum theory AND - Makes an informed judgment implicitly or explicitly about Einstein's contribution to quantum theory	3
Demonstrates some understanding of the contribution of Einstein to quantum theory	2
Identifies a contribution of Einstein to modern physics	1

Sample answer:

Einstein contributed a great deal to Quantum Theory. Until this time in history, classical theory was the prevailing theory and it was the physics of the large scale. Quantum theory deals with the physics of the small. Planck had proposed that light energy came in packets or quanta when he was investigating black body radiation. The energy of the photon is given by $E = hf$. Einstein was able to use Planck's ideas and apply them to both black body radiation and the photoelectric effect to formulate explanations for how and why they work. Einstein stated that light comes in quanta called photons, which have an energy ($E = hf$), and when this photon interacts with electrons in a metal surface, it gives all or none of its energy to the electron.

Question 29 (a)

Criteria	Marks
• Correct substitution into relevant equation to calculate the force required	2
• Identifies one correct step in calculating the force required	1

Sample answer:

$$F = mg = 0.05 \times 9.8 = 0.49 \text{ N}$$

Question 29 (b)

Criteria	Marks
- Equates two torque equations AND - Shows correct substitution into both equations to calculate the current required	3
- Equates two torque equations OR - Shows correct substitution into ONE torque equation	2
• Identifies a correct step in calculating the current required	1

Sample answer:

$$\tau = Fd = nBIA \cos \theta$$

$$mgd = nBIA \cos 0^\circ$$

$$0.05 \times 9.8 \times 0.004 = 100 \times 0.1 \times I \times 0.0012 \times 1$$

$$1.96 \times 10^{-3} = 0.012 I$$

$$I = 0.16 \text{ A}$$

Question 30 (a)

Criteria	Marks
- Provides a correct explanation - Identifies centripetal force towards centre and relates this force to simulation of gravity	2
Identifies a force is acting on the astronaut	1

Sample answer:

The rotating wheel provides a centripetal force (acceleration) towards the centre. The astronaut reacts against this force according to Newton's 3rd Law, thus providing a simulated gravity.

Question 30 (b)

Criteria	Marks
Demonstrates correct process to determine the rotational speed	2
Shows an understanding in deriving the orbital velocity formula	1

Sample answer:

$$\begin{aligned}
 F &= \frac{mv^2}{r} = mg \\
 g &= \frac{v^2}{r} \\
 v^2 &= gr \\
 v &= \sqrt{gr} \\
 &= \sqrt{9.8 \times \frac{550}{2}} \\
 &= 51.9 \text{ m s}^{-1}
 \end{aligned}$$

Question 31

Criteria	Marks
- Shows thorough understanding of the changes that occur to the electrical properties of materials - Clearly relates changes in electrical properties of materials to the development of new technologies - Shows good understanding of the benefits of these technologies to society - Explains in a coherent and logical manner	7
- Shows good understanding of the changes that occur to the electrical properties of materials - Relates changes in electrical properties of materials to the development of new technologies - Shows good understanding of the benefits of these technologies to society	6
- Shows good understanding of changing electrical properties of materials - Relates the change(s) to the development of a new technology - Shows sound understanding of the benefits of this technology to society OR - Shows sound understanding of electrical properties of materials - Links these electrical properties to relevant new technologies - Shows some understanding of the benefits of these technologies to society	5
- Shows a sound understanding of electrical properties of materials - Links the electrical properties/property to the development of a relevant new technology - Shows sound understanding of the benefits of this technology to society OR - Shows some understanding of electrical properties of materials - Identifies relevant new technologies - Shows some understanding of the benefits of these technologies to society	4
- Describes a relevant new technology and identifies a benefit of this technology to society OR - Links electrical properties of materials to the development of technologies/benefits to society OR - Shows sound understanding of electrical properties of materials	3
Shows some understanding of electrical properties and/or relevant new technologies and/or relevant benefits to society	2
Identifies a relevant piece of information on electrical properties / new technologies / benefits to society	1

Sample Answer:

Changes in the electrical properties of materials can produce semi-conduction materials or superconductors.

P and N type semi conductors can then be used to make components such as transistors and solar cells.

Transistors have led to the development of miniaturised computers, phones etc rather than those possible with thermionic devices. These items have benefited society by making an information rich society that has efficient communication.

Mobile phones benefit society because people can now communicate from most places. This can save lives allowing a rapid response in car accidents.

Solar cells allow for electricity to be produced and delivered in remote areas and is a clean form of energy which reduces carbon emissions and ultimately improve society's health.

Super conductors allow a much better way to detect small magnetic fields, which can be used in SQUIDS for medical applications to improve diagnosis and hence the health of the population.

Superconductors can be used in sending and making electricity, reducing energy losses and hence reducing carbon emissions and greenhouse effect.

Answers could include:

- Mentions multiple electrical properties – eg conductors, semiconductors and superconductors
- Mentions multiple technologies – eg transistors, microprocessors, Maglev trains, MRI, electricity supply
- Mentions multiple benefits to society – eg reduced costs, reduction of electricity use, increase in portability (miniaturisation), employment, communications, globalisation, processing computer speed, delocalised workforce, improved health, improved medical diagnosis, etc.

Section II

Question 32 – Geophysics

Question 32 (a)

Criteria	Marks
Correctly describes how a relevant piece of equipment detects seismic waves	3
Provides a general description of a relevant piece of equipment	2
- Names a piece of relevant equipment OR - Identifies a relevant feature	1

Sample answer:

The seismometer works on the principle of inertia of one component of the device measuring movement of another component of the device that is attached to Earth. A freely suspended inertia mass is attached to a base on the Earth. As the Earth shakes the mass remains stationary. This is connected to a needle which imparts a graphical representation onto a drum.

Question 32 (b) (i)

Criteria	Marks
Describes a plausible investigation procedure including apparatus	3
Outlines the main steps of a suitable procedure which may not include apparatus	2
- Lists relevant equipment OR - Identifies some steps	1

Answers could include:

Procedure

- Measure mass of rock sample using balance
- Partially fill measuring cylinder with water. Record volume
- Tie string to rock, immerse completely in water in measuring cylinder
- Record new volume
- Difference between the two volumes equals the volume of the rock
- Use $D = \frac{M}{V}$ to calculate density of rock
- Repeat for each different rock type

Question 32 (b) (ii)

Criteria	Marks
• Explains how the reliability of results could be determined	2
• Shows a basic understanding of the results of the procedure	1

Sample answer:

The reliability of the results can be determined by comparing density of rocks obtained from repetitions of the experiment to see if they are consistent.

Question 32 (c) (i)

Criteria	Marks
• Explains by relating activity to hot spot and to how the hot spot is formed	2
• Shows a basic understanding of the volcanic activity	1

Sample answer:

The volcanic activity on the island of Hawaii is associated with a hot spot on the ocean floor where hot solid material is rising in narrow cylindrical jets from deep within the mantle.

Question 32 (c) (ii)

Criteria	Marks
• Provides an explanation by relating the position of the islands to moving plates and relating this to theory of plate tectonics	3
• Shows some understanding of plate tectonics and/or the formation of the island chain	2
• Provides one piece of relevant information	1

Sample answer:

Although the Hawaiian Islands are in the middle of the Pacific plate they do provide evidence of plate movement. The plate moved to the NW but the hot spot remained stationary. Therefore the further NW of Hawaii the older and more stable the islands, eg Kauai was on the hot spot 3.8 to 5.6 million years ago and as the plate moved NW it moved off the hot spot and stopped volcanic activity. The next island formed over the hot spot was Oahu 2.2 to 3.4 million years ago. Therefore the island chain provides evidence that the plate moves not just at the boundaries.

Question 32 (d)

Criteria	Marks
Explains the significance of the shape of the gravity anomaly line by relating the density of rock and the gravity anomaly to the topographic map	3
Relates gravity anomaly to density of underlying rock OR to topographic map	2
States an understanding of gravity anomaly	1

Sample answer:

The gravity anomaly (Bauger Anomaly) shows density variations due to differing density of rocks below the datum without the effects of topography or latitude.

Large gravity lows (troughs) are associated with accumulation of lighter, less dense material near trenches and long gravity high (crest) due to cold dense slabs of lithosphere descending from beneath the volcanic zone.

Question 32 (e)

Criteria	Marks
Shows correct process to determine the acceleration	3
- Derives correct equation for determining acceleration OR - Equates two relevant equations	2
Partial substitution into a relevant equation	1

Sample answer:

$$F_g = \frac{G m_o m_E}{r_E^2} \quad F_g = m_o g$$

$$m_o g = \frac{G m_o m_E}{r_E^2}$$

$$g = \frac{G m_E}{r_E^2}$$

$$g = \frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{(6357 \times 10^3)^2}$$

$$g = 9.9 \text{ ms}^{-2}$$

Question 32 (f)

Criteria	Marks
- Identifies major issues and provides points for and/or against the effectiveness of using geophysical methods in mineral exploration AND environmental monitoring - Shows thorough understanding of geophysical methods - Uses clear and scientifically correct language	6
- Identifies issues and provides points for and/or against the effectiveness of using geophysical methods in mineral exploration AND environmental monitoring - Shows good understanding of geophysical methods	5
- Outlines issues in relation to the effectiveness of using geophysical methods in mineral exploration and/or environmental monitoring - Shows sound understanding of geophysical methods	4
Outlines issues in relation to the use of geophysical method(s) in mineral exploration and/or environmental monitoring	3
Shows some understanding of geophysical method(s) and/or mineral exploration and/or environmental monitoring	2
Shows a basic understanding of geophysical method/mineral exploration/ environmental monitoring	1

Sample Answer:

Geophysical methods have proven to be very beneficial in both mineral exploration and environmental monitoring. The main geophysical methods employed include gravity methods, seismic methods, magnetic methods, electrical methods, radiometric methods and geothermal methods.

Seismic methods are capable of looking into the earth without the need for removal of the surface rocks. By using artificially generated sound waves and their reflection from subsurface layers they are able to detect the gross geological features of an area.

The use of seismic methods have enabled the presence and boundaries of water, gas and oil deposits to be accurately determined, reducing the large scale need for drilling and excavation of the landscape in search of the deposits.

Seismographic stations all over the world can detect earthquakes. The use of this network of stations for early detection of earthquake wake could reduce the fatalities that occur as a result of earthquakes.

Magnetic methods used in geophysics work on the magnetic susceptibility of minerals. The magnetic susceptibility depends on the amount of ferromagnetic minerals present in the rock. Magnetic prospecting can be done using high-resolution airborne magnetic surveys where a magnetometer is towed behind an airplane detecting the magnetic field from the terrain below. Satellites are also used for that purpose. This process is often non-invasive, hence causes no damage to the vegetation. The extent and location of large ferromagnetic ore deposits can be detected and planning of mining infrastructure and management can be done without adverse impact on the environment. Magnetic methods have enabled the monitoring of the earth's

magnetic field as well as monitoring of magnetic anomalies associated with the spreading of the ocean floor.

Answers could include:

Environmental monitoring can vary from the local to regional and even national scale with instruments ranging from hand held to high tech satellites. The data collected can be stored in digital format so it can be reprocessed so that comparisons can be made and environmental monitoring can be continuous.

Question 33 — Medical Physics

Question 33 (a)

Criteria	Marks
• Shows a comprehensive understanding of differences between X-ray images and CAT scans	3
• Clearly shows the differences between X-ray images and CAT scans	2
• Identifies a feature of X-ray images or CAT scans	1

Sample answer:

<i>X-ray images</i>	<i>CAT scans</i>
• Fewer shades of grey in image (30 shades) • One 2D plane image formed at a time • X-ray images show bone early	• More shades of grey in image (256 shades) • Helical scanning forms multiple images in all 3 planes (x, y and z) which can be formed in 3D image • CT images show soft tissue and bone structure clearly

Question 33 (b)

Criteria	Marks
• Outlines a first-hand investigation to demonstrate the transfer of light by optical fibres	2
• Identifies ONE relevant feature of the process	1

Sample answer:

A light beam, either single ray or laser pointer, is sent into a glass rod at an angle to the surface. The light beam can be seen in a zig zag pattern down the rod. This is called total internal reflection, as shown in the diagram below.



Question 33 (c)

Criteria	Marks
Correctly explains how tissue samples of internal organs can be obtained using an endoscope	3
Shows some understanding of the use of an endoscope to obtain tissue samples	2
Identifies a piece of relevant equipment or a step in the process	1

Sample answer:

Tissue samples can be taken through keyhole surgery using an endoscope. Along with optical fibres to illuminate and view inside the body, endoscopes have attachments that can cut or snip small tissue samples and an extract that can remove the sample from the patient.

Question 33 (d) (i)

Criteria	Marks
Identifies changes in orientation of nuclei magnetic axis due to the application of strong magnetic field	2
Identifies orientation of nuclei magnetic axis due to the strong magnetic field	1

Sample answer:

Nuclei with net spin have a magnetic moment that has a random orientation when no external magnetic field is applied. After a strong magnetic field is applied, the magnetic moment of the nuclei line up either parallel or anti-parallel to the external magnetic field.

Question 33 (d) (ii)

Criteria	Marks
Correctly explains the difference in the relaxation times between hydrogen in water and hydrogen in other molecules	3
Shows an understanding of the relaxation time of hydrogen in water and that of hydrogen in other molecules	2
Identifies some relevant information	1

Sample answer:

Hydrogen in water has a longer relaxation time because the interaction with its surroundings (water) is small. The relaxation time of hydrogen in other molecules is shorter because it has greater interactions with its surroundings.

Question 33 (e) (i)

Criteria	Marks
Describes how the Doppler effect is used in ultrasonics to investigate blood flow characteristics	3
Outlines how the Doppler effect is used in ultrasonics	2
Shows a basic understanding of the Doppler effect	1

Sample answer:

The Doppler Effect is the apparent change in frequency of a sound wave that is observed when there is relative movement between the source of the sound and the observer. In ultrasonics, the ultrasound waves are sent out from the transducer towards the moving blood cells and the reflected ultrasound waves are detected by the transducer. The change in the frequency of the ultrasound waves observed is used to determine the direction and speed of the blood cells. This can assist in diagnosis of blockages or leakages in blood vessels, especially in the heart.

Question 33 (e) (ii)

Criteria	Marks
• Outlines TWO technologies required for an ultrasound scan	3
• Outlines ONE technology required for an ultrasound scan	2
• Identifies TWO technologies required for an ultrasound scan	1

Sample answer:

For Doppler ultrasound to be used in an ultrasound scan, both the cathode ray tube monitor and the computer needed to be developed. The processing speed of computers has enabled the formation of an ultrasound image and calculation of direction and speed of blood flow to happen in a relatively short period of time.

Question 33 (f)

Criteria	Marks
- Identifies both gamma scans and PET as diagnostic tools using radioactive isotopes - Identifies radioactive isotopes for gamma scans and PET from list provided - Fully justifies the suitability of the identified radioactive isotopes for use in diagnostic imaging techniques - Shows a thorough understanding of both gamma scans and PET and their use as a diagnostic tool with the radioactive isotopes - Uses clear and scientifically correct language	6
- Identifies both gamma scans and PET as diagnostic tools using radioactive isotopes - Identifies radioactive isotopes for gamma scans and PET from list provided - Provides a good justification for the suitability of the identified radioactive isotopes for use in diagnostic imaging techniques - Shows a good understanding of both gamma scans and PET and their use as a diagnostic tool	5
- Identifies both gamma scans and PET as diagnostic tools using radioactive isotopes - Identifies radioactive isotopes for gamma scans and PET from list provided - Provides some justification for the suitability of the identified radioactive isotopes for use in diagnostic imaging techniques - Shows a sound understanding of either gamma scans or PET and their use as a diagnostic tool	4
- Identifies either gamma scans or PET as diagnostic tools using radioactive isotopes - Identifies a radioactive isotope for either gamma scans or PET from list provided - Gives reason(s) for the suitability of the identified radioactive isotope for use in diagnostic tools - Shows some understanding of either gamma scans or PET and their use as a diagnostic tool	3
Shows some understanding of diagnostic tool(s) and/or radioactive isotope(s)	2
Shows a basic understanding of diagnostic tool/radioactive isotope	1

Sample answer:

Radioactive isotopes are used to show functioning of areas of interest in the body. Radioactive isotopes are attached to pharmaceuticals that roam the body and position themselves in certain areas. Their radioactive decay products are detected with diagnostic imaging tools such as Gamma scan and PET devices.

Radioactive isotopes used for diagnostics are short-lived so that they can decay fast and don't harm the body, yet radiate long enough to be transported and given to the patient. Fluorine-18 and Technetium-99m are such isotopes. T-99m is a gamma emitter and thus suitable for gamma scans. F-18 is a β^+ or positron emitter and thus suitable for PET scans.

The gamma rays emitted by T-99m pass through the body without harming it. The uptake of T-99m is determined by metabolism of the organ and hence different concentration of gamma rays can be detected outside the body by gamma cameras. A computer is used to further process gamma ray locations and concentration.

The F-18 emitted positron combines with a nearby electron, together they annihilate to produce 2 gamma rays which are emitted in opposite direction out of the body. This allows a gamma detector to locate the emission region with high accuracy and hence high-resolution location specific images can be produced for diagnostic purpose.

Question 34 — Astrophysics**Question 34 (a)**

Criteria	Marks
• Outlines a suitable method • Outlines the results • Links the results to desirability in terms of sensitivity	3
• Gives features of a suitable method	2
• Identifies a feature of a suitable method	1

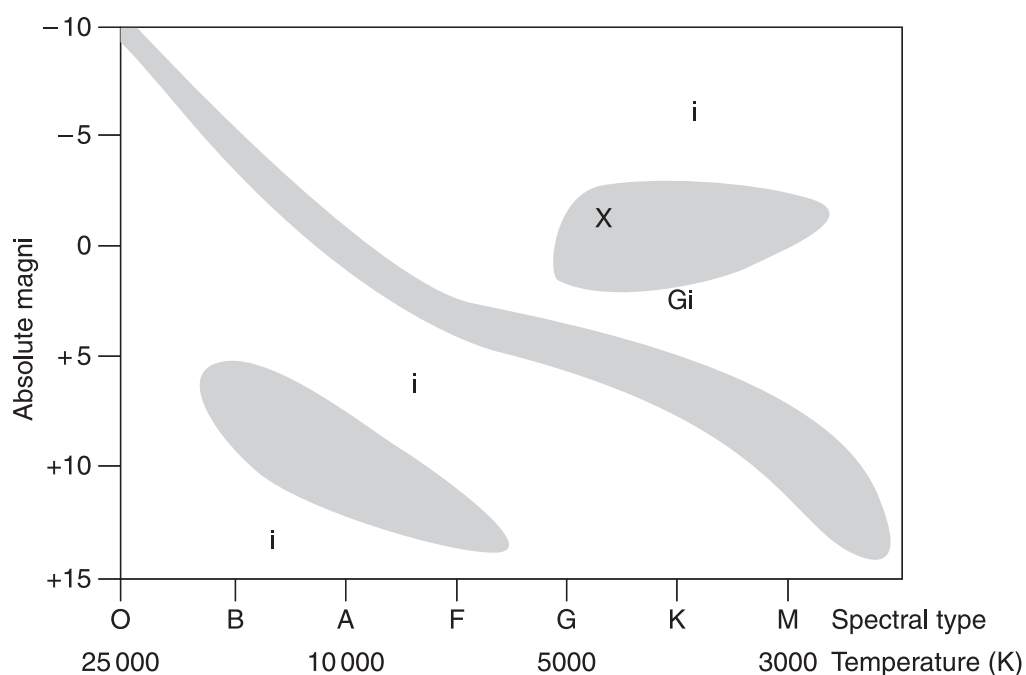
Sample answer:

An investigation can be performed where light is collected by a photo meter through different sized lenses. The source of light is the sun. The investigation should show that the larger the diameter of the lens the more light is collected due to the larger surface area. The telescope becomes more sensitive as more light is collected. Hence, it is desirable in terms of sensitivity for a telescope to have a larger diameter lens.

Question 34 (b) (i)

Criteria	Marks
- Correctly places and labels Matar on the HR diagram - Draws and labels an HR diagram showing the main star groupings	3
- Correctly places Matar on the HR diagram - Draws an HR diagram	2
Shows some understanding of an HR diagram	1

Sample answer:



Question 34 (b) (ii)

Criteria	Marks
Shows a good understanding of the differences between the two stars	2
Identifies one difference between the stars other than mass	1

Sample answer:

Matar is more luminous and larger in radius than the sun. It is a giant and hence is less dense than the sun.

Question 34 (c)

Criteria	Marks
• Describes in detail the CNO Cycle	3
• Provides some features of the CNO cycle	2
• Identifies a feature of the nuclear fusion reaction	1

Sample answer:

In an A2 main sequence star the CNO cycle will be the dominant fusion reaction in the core of the star. This reaction takes place at higher temperatures than the p p reaction and as it uses carbon nitrogen and oxygen as a nuclear catalyst the reaction is faster than the p p reaction. In the overall reaction 4 hydrogens are converted into helium releasing two positrons and a neutrino in a multi-step reaction. Carbon, oxygen and nitrogen are intermediary products.

Question 34 (d)

Criteria	Marks
• Describes key physical processes and nuclear reactions during the evolution	3
• Outlines the transition process OR	2
• Shows some understanding of the features of the transition process	
• Identifies features of the transition process	1

Sample answer:

When a main sequence star such as our sun reaches the end of its life hydrogen in the core is depleted. The nuclear reaction is no longer sustained and the core collapses heating up the surrounding shell and the core. Hydrogen in the shell will eventually begin to fuse to helium. The surrounding shell expands due to the increased radiation pressure and as it does it cools becoming more red in colour. The helium fusion reaction in the core will start suddenly and produce a 'helium flash' the radius of the star will decrease and the star could become unstable and pulsate as a periodic variable.

Question 34 (e) (i)

Criteria	Marks
Correct substitution to determine the distance and compares to the stated value	3
- Correct substitution to determine the distance OR - Partially substitutes into distance modulus formula and compares the distance calculated to the stated value	2
- Finds the absolute magnitude OR - Shows an understanding that the distance modulus needs to be used	1

Sample answer:

Absolute magnitude $M = -4$

The distance modulus is $m - M = 6.5 - (-4) = 10.5$, using the distance modulus formula this corresponds to a distance of 1260 pc which is the claimed value.

$$M = m - 5 \log \left[\frac{d}{10} \right]$$

$$\begin{aligned}
 m - M &= 5 \log \left[\frac{1260}{10} \right] \\
 &= 10.5
 \end{aligned}$$

Question 34 (e) (ii)

Criteria	Marks
States one advantage and one disadvantage	2
States one advantage or one disadvantage	1

Sample answer:

Disadvantage: The method is not that accurate

Advantage: It can be done to predict the distance to a cluster containing a Cepheid variable

Question 34 (f)

Criteria	Marks
• Demonstrates coherence and logical progression and includes correct use of scientific ideas and principles • Describes methods and relates the methods to the physics principles behind the determination of standard properties • Uses information in the table to express their answer • Uses clear and scientifically correct language	6
• Describes methods and physics principles behind the determination of standard properties • Refers to information in the table	5
• Describes methods and physics principles behind the determination of some standard properties OR • Outlines methods and physics principles behind the determination of standard properties	4
• Outlines method(s) and/or physics principle(s) for determining standard properties	3
• Shows some understanding of physics principle(s) and/or method(s) for determining standard property	2
• Shows a basic understanding of a physics principle/method for determining standard property	1

Sample answer:

To determine the characteristics of a star the visible light can be broken up with a spectrograph and then examined. The spectrum is compared to known spectra of other stars. Stars of the A class will have absorption lines showing strong hydrogen lines and ionised metal lines in the spectra. By comparing radiation emitted by the star to a black body radiation curve the colour of the star can also be determined as the combination of wavelengths emitted will be white. The temperature of the star can also be determined in this way as stars act as black body radiators and will produce a specific radiation curve at a specific temperature in this case between 7500–10000 K. The mass of a star can be determined by assuming that stars of the same spectral class and same luminosity class will be the same mass. The standard is produced by determining the mass of a binary star using Kepler's law. The separation of the stars can be determined and by collecting the light a CCD/computer or similar device analyses the amount of light to determine the period of the variation in light which will be the period of orbit as the stars rotate about a common point from an eclipsing binary. $m_1 + m_2 = 4\pi^2 r^3 / GT^2$. The luminosity or absolute magnitude can be determined by using photometry eg taking a photo and measuring the magnitude to determine the apparent magnitude and if the distance to the star is determined then the distance modulus formula can be used to determine the absolute magnitude or luminosity. Distances can be determined by spectroscopic parallax or trigonometric parallax depending on the distance to the star. Both mass and luminosity are compared to that of the Sun.

Question 35 — From Quanta to Quarks

Question 35 (a) (i)

Criteria	Marks
• Outlines key features of the Rutherford model	2
• States a feature of the Rutherford model	1

Sample answer:

A small, dense, positively charged nucleus, with electrons orbiting around it at a distance. The atom is mostly empty space.

Question 35 (a) (ii)

Criteria	Marks
• Identifies one limitation of the Rutherford model and explains the relevant Bohr's postulate to account for the limitation	3
• Identifies one limitation of the Rutherford model or identifies relevant Bohr's postulate to account for a limitation	2
• Identifies a feature of the Bohr model	1

Sample answer:

Rutherford's model could not explain the hydrogen emission lines.

Bohr's model placed electrons into quantised energy shells. When electrons moved from a higher to a lower shell, they released energy. This energy corresponded to the frequencies in the emission spectra.

Question 35 (b) (i)

Criteria	Marks
• Shows the correct process to determine the energy released	3
• Shows some understanding of calculating the energy released	2
• Identifies that there is a mass difference in reactants and products	1

Sample answer:

$$U + n = 235.0439 + 1.008665$$

$$\text{Reactants} = 236.0526 \text{ amu}$$

$$Ba + Kr + (3 \times n) = 140.9144 + 91.9263 + (3 \times 1.008665)$$

$$\text{Products} = 235.8433 \text{ amu}$$

$$\text{Mass defect} = 236.0526 - 235.8433$$

$$\text{Mass defect} = 0.2093 \text{ amu}$$

$$\text{Energy} = 0.2093 \times 931$$

$$\underline{\text{Energy} = 194.8583 \text{ MeV}}$$

Question 35 (b) (ii)

Criteria	Marks
• Explains the production of energy by applying Einstein's mass/energy equivalence	2
• Identifies that the mass has been converted into energy	1

Sample answer:

Because there is a mass difference between the products and the reactants, this mass is converted into the released energy, as mass and energy are equivalent as stated by Einstein.

Question 35 (c)

Criteria	Marks
Identifies the features of a neutrino AND explains why it was suggested AND why Pauli did not 'discover' the particle	4
- Identifies the features of a neutrino AND - Shows some understanding of why it was 'suggested' but not discovered by Pauli	3
- Identifies the features of a neutrino OR - Shows some understanding of why it was 'suggested' but not discovered by Pauli	2
Identifies a feature of the neutrino	1

Sample answer:

Pauli is said to have suggested the new particle to explain why there were varying energies in the emitted electrons during β decay. This appeared to break the conservation laws. Pauli suggested the neutrino carried the 'missing energy' but was massless and therefore undetectable. His suggested particle solved the problem of the conservation laws, but he had no evidence that it existed, hence why it was suggested but not discovered by Pauli.

Question 35 (d) (i)

Criteria	Marks
• Identifies all the forces proposed by the standard model of the atom	2
• Identifies TWO forces proposed by the standard model of the atom	1

Sample answer:

Forces	
Electromagnetic	Gravity
Strong Force	Weak Force

Question 35 (d) (ii)

Criteria	Marks
• Explains how the stability of atomic nuclei is maintained	3
• Provides some relevant features	2
• Provides a relevant feature	1

Sample answer:

The nuclei of atoms are stable due to the presence of the strong nuclear force. This force is an attractive force that holds the nucleons together. It is stronger than the repulsive electrostatic force that exists between the protons.

Question 35 (e)

Criteria	Marks
- Shows a thorough understanding of the contributions made by de Broglie - Describes how de Broglie used existing concepts and ideas to come up with new interpretations - Uses clear and scientifically correct language	6
- Shows a good understanding of the contributions made by de Broglie - Outlines how de Broglie used existing concepts/ideas to come up with new interpretations	5
- Shows a sound understanding of the contributions made by de Broglie - Identifies existing concepts/ideas de Broglie used to come up with new interpretations in our understanding of the structure of the matter	4
Shows some understanding of the contributions made by de Broglie	3
Identifies some contribution(s) made by de Broglie and/or existing concept(s) and/or idea(s)	2
Identifies a contribution made by de Broglie or existing concept or idea	1

Sample answer:

De Broglie was able to utilise the current understanding of matter to further develop our understanding of the structure of matter. At the time there was an understanding of the nature of light, which was energy, displaying particle-like properties. Also, a model of the atom was established displaying quantised electron energy levels, but this was yet to be explained. De Broglie utilised the ideas about light being a particle and applied this to electrons, stating that it could behave as a wave. By establishing that electrons exist in stationary states that correspond to the wavelength of the electron, and that they could only move between energy levels that correspond to this wavelength, de Broglie was able to provide an explanation for the stability and quantised nature of the electrons in the atom. De Broglie was then able to describe the wave particle duality of matter. In this way de Broglie increased our understanding of the nature and structure of matter.

Question 36 — The Age of Silicon

Question 36 (a)

Criteria	Marks
Shows correct process to determine the voltage	3
- Determines that the LDR resistance at 20 lux is 30 kΩ OR - Partial substitution into relevant equation	2
- Shows a basic understanding of interpreting the graph OR - Identifies some relevant data and formula	1

Sample answer:

From graph 20 lux → 30 kΩ

$$\frac{5V \times 30k\Omega}{(30k\Omega + 10k\Omega)} = 3.75V$$

Question 36 (b)

Criteria	Marks
- Describes how output voltage produced by a LDR in a circuit relates to light level and can be used by a camera's electronics - Identifies that this setup would be used for light metering to produce correct exposure	3
Identifies relevant features	2
Identifies a relevant feature	1

Sample answer:

The light level falling on the LDR is converted to a voltage. This voltage is then sent to the camera's electronics systems to control the exposure via shutter speed/aperture. This allows for the right amount of light to fall on the film (or CCD) to produce a correct exposure. Thus the LDR circuit acts as a light meter.

Question 36 (c)

Criteria	Marks
Explains how relevant input and output transducers interact to control the temperature in a room	4
- Shows some understanding of how relevant input and output transducers can be used OR - Shows some understanding of how a relevant transducer can be used - Demonstrates an understanding of the difference between electronic and electric circuits	3
Shows some understanding of how a relevant transducer can be used	2
Identifies one relevant feature	1

Sample answer:

A thermistor placed in a room acts as an input transducer to monitor temperature by converting temperature to a voltage in a voltage divider. Through an electronic circuit, an output voltage would be used to switch on and off the heating lamp, but since the output voltage is too small to operate a heating lamp, a relay must be used. The relay's input is the low voltage from the electronic circuit, and the output switches the heating lamp on or off.

Question 36 (d)

Criteria	Marks
Draws a correct truth table	3
Draws a partially correct truth table	2
Shows some understanding of a truth table	1

Sample answer:

A	B	C	Output
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Question 36 (e) (i)

Criteria	Marks
• Correctly describes an ideal operational amplifier	2
• Identifies a feature of an ideal operational amplifier	1

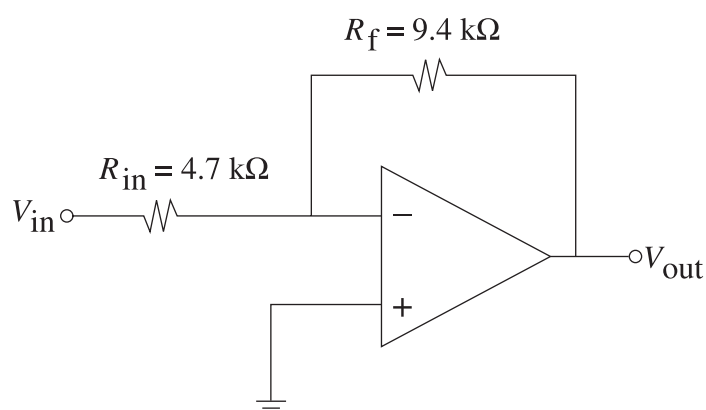
Sample answer:

An ideal operational amplifier outputs a reproduction of its input signal, only larger in voltage. It has infinite input impedance, zero output impedance, infinite open-loop gain and infinite bandwidth.

Question 36 (e) (ii)

Criteria	Marks
• A substantially correct diagram with the $9.4\text{ k}\Omega$ resistor drawn in correct location	2
• Either correct location OR correct value resistor drawn	1

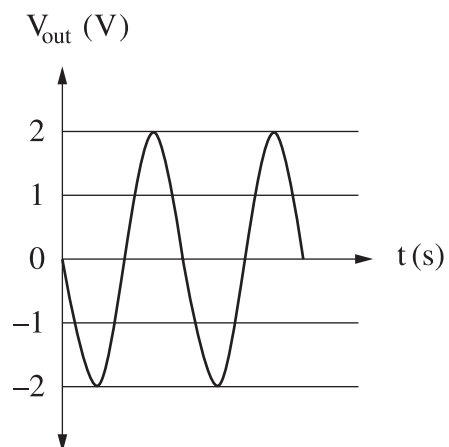
Sample answer:



Question 36 (e) (iii)

Criteria	Marks
• Correct shape, inverted and correct amplitude	2
• Correct shape, inverted or correct amplitude	1

Sample answer:



Question 36 (f)

Criteria	Marks
• Demonstrates thorough knowledge and understanding of the development of electronics, including past developments and future limitations • Clearly relates the above to society's increasing computer needs • Makes an informed judgement implicitly or explicitly about how the development of electronics has met the increasing computing needs of society • Uses clear and scientifically correct language	6
• Demonstrates good knowledge and understanding of the development of electronics, including past developments and limitations • Some relation of the above to society's increasing computer needs	5
• Demonstrates sound knowledge and understanding of the development of electronics, including past developments or limitations • Some relation of the above to society's increasing computer needs	4
• Demonstrates some knowledge and understanding of the development of electronics, including past developments or limitations	3
• Demonstrates an understanding of the development of electronics or increasing computer needs	2
• Identifies a relevant fact	1

Sample answer:

Early computers each employed hundreds of thousands of single transistors. However, as the need for faster, more powerful and lightweight computers arose, scientists had to work out a way of packing more transistors into a small space. This led to the integrated circuit, which contained millions of transistors on a single chip. This uses far less power and takes up far less space for a given processing ability.

As computing needs increase, more and more transistors are placed onto a single chip. This means they become closer and closer to physical limits due to the size of atoms and so quantum effects become significant. This is where electrons tunnel across the transistors and interfere with one another. As a result, computers have had to be designed differently to overcome these effects as this limit has been reached. Portable computers now make use of multiple core processors working in parallel to increase computing power rather than further miniaturisation of the chip.

From these developments, society has access to smaller and more powerful computers, increasing productivity and communication between individuals and businesses worldwide. It has also led to changes in the way people are entertained by music and video through the use of mp3 players and videos on their portable devices. Portable GPS devices have made it possible to navigate easily in your car without stopping to look at a map. There have also been negative impacts as people tend to rely so much on the technology, that when technology fails for one reason or another, communication breaks down.

Physics

2013 HSC Examination Mapping Grid

Section I Part A

Question	Marks	Content	Syllabus outcomes
1	1	9.1.11.2a, 9.3.2.3.2	H11
2	1	9.2.3.2.4	H7, H9
3	1	9.3.1.2.4, 9.3.1.2.5	H9, H6
4	1	9.1.11.3b, 9.2.2.3.2	H11
5	1	9.4.2.2.2	H2
6	1	9.2.2.2.2, 9.2.2.2.1	H6, H9
7	1	9.3.4.3.3	H7, H9
8	1	9.2.4.2.4	H6
9	1	9.2.2.2.12	H6, H7, H9
10	1	9.3.4.3.4	H9
11	1	9.4.4.2.1	H11
12	1	9.4.4.3.3	H9
13	1	9.3.2.2.3	H9
14	1	9.4.1.2.7	H9
15	1	9.3.1.3.3, 9.3.1.3.4	H7, H9, H10
16	1	9.4.3.2.2, 9.4.3.2.1	H7, H9, H10
17	1	9.3.1.3.4, 9.3.2.2.5	H9
18	1	9.1.14.1c, 9.2.4.2.2	H14
19	1	9.2.4.3.5	H6
20	1	9.4.2.2.5, 9.4.2.2.6	H10

Section I Part B

Question	Marks	Content	Syllabus outcomes
21 (a)	2	9.2.2.3.3	H1
21 (b)	2	9.1.12.4e, 9.2.2.3.3	H12.4e
22	3	9.2.1.3.2	H6, H13.1e, f, g, H14.1a
23 (a)	2	9.2.2.2.10	H6
23 (b)	3	9.2.2.3.5	H6
24	5	9.4.2.3.3	H3
25 (a)	1	9.3.1.2.2	H9
25 (b)	3	9.3.1.3.1	H9
26 (a)	2	9.4.1.3.3	H9
26 (b)	3	9.4.1.2.7, 9.4.1.3.3	H9
27 (a)	3	9.3.2.2.4	H7
27 (b)	4	9.3.3.2.2, 9.3.5.3.2	H7

Question	Marks	Content	Syllabus outcomes
28	6	9.4.2.3.2	H1, H2, H10
29 (a)	2	9.2.1.3.3	H9
29 (b)	3	9.3.1.3.4, 9.3.1.2.3	H9
30 (a)	2	9.2.2.2.8	H6, H9
30 (b)	2	9.2.2.2.5, 9.2.2.3.4	H6, H9
31	7	9.4.3.2.6, 9.4.3.3.2., 9.4.3.3.3, 9.4.3.2.8, 9.4.4.3.4, 9.4.4.3.5	H3, H4, H9

Section II

Question	Marks	Content	Syllabus outcomes
Question 32		Geophysics	
(a)	3	9.5.3.2.4	H2, H3, H8
(b) (i)	3	9.5.1.3.1	H11, H12, H13, H14
(b) (ii)	2	9.5.1.3.1	H11, H12, H13, H14
(c) (i)	2	9.5.4.2.3	H7
(c) (ii)	3	9.5.4.2.3	H1, H2, H4, H7
(d)	3	9.5.2.1.4, 9.5.2.1	H9
(e)	3	9.5.2.3.3	H6, H9
(f)	6	9.5.1.3.1, 9.5.5.1.2	H13, H14
Question 33		Medical Physics	
(a)	3	9.6.2.3.2	H8, H10
(b)	2	9.6.2.3.3, 9.6.2.2.5	H12
(c)	3	9.6.2.3.3, 9.6.2.2.7	H14
(d) (i)	2	9.6.4.2.4	H7, H9, H10
(d) (ii)	3	9.6.4.2.8	H7, H9, H10
(e) (i)	3	9.6.1.2.8	H6, H7
(e) (ii)	3	9.6.1.2.4, 9.6.1.2.8	H3, H7, H9, H10
(f)	6	9.6.3.2.1, 9.6.3.2.2, 9.6.3.2.3, 9.6.3.2.4, 9.6.3.2.5	H3, H7, H8, H9, H10, H14

Question 34		Astrophysics	
(a)	3	9.7.1.3.1	H3, H8
(b) (i)	3	9.7.6.3.2	H10
(b) (ii)	2	9.7.6.3.2, 9.7.3.2.4	H8
(c)	3	9.7.6.3.2, 9.7.6.3.3, 9.7.6.3.4	H7
(d)	3	9.7.6.2.2, 9.7.6.3.3, 9.7.6.2.6	H7, H9
(e) (i)	3	9.7.4.3.1, 9.7.5.2.4	H8, H10
(e) (ii)	2	9.7.5.2.4	H10
(f)	6	9.7.5.2.2, 9.7.5.3.2, 9.7.4.2.5, 9.7.3.2.2, 9.7.3.2.4, 9.7.3.2.5, 9.7.3.2.1	H8, H10
Question 35		From Quanta to Quarks	
(a) (i)	2	9.8.1.2.1	H1, H10
(a) (ii)	3	9.8.1.2.3	H2, H10
(b) (i)	3	9.8.3.2.9	H6
(b) (ii)	2	9.8.3.3.2	H10
(c)	4	9.8.3.2.6	H1, H6
(d) (i)	2	9.8.4.2.5	H9, H10
(d) (ii)	3	9.8.4.2.5	H10
(e)	6	9.8.2.2.1, 9.8.2.2.4, 9.8.2.2.3	H2, H10
Question 36		The Age of Silicon	
(a)	3	9.9.2.3.3	H13
(b)	3	9.9.3.2.3	H3
(c)	4	9.9.4.2.1, 9.9.3.2.6	H7
(d)	3	9.9.5.2.1, 9.9.5.3.2	H9
(e)(i)	2	9.9.6.2.1	H3
(e)(ii)	2	9.9.6.2.7, 9.9.6.2.8, 9.9.6.3.3	H13
(e)(iii)	2	9.9.6.3.2	H13
(f)	6	9.9.1.2.1, 9.9.1.2.2, 9.9.1.2.3, 9.9.1.3.1, 9.9.7.2.1, 9.9.7.2.2, 9.9.7.2.3	H1, H3, H4